

ROOTMESH · DESIGN DECISION RECORD

RootMesh recommendations accepted

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DOCUMENT CONTROL

Document ID	RMS-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/rootmesh		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1 and P1 decided by Amish on 2026-10-02 as recommended
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of the remaining review recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items A1 to A4; items O1 and P1 decided on 2026-10-02 as recommended (Amish, 2026-10-02: "i approve your recommendations for all 555 open decisions.")

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." The TRL 2 review items that carried a recommendation (D1 to D9) had already been decided in RMS-DDR-001 and are unchanged. This record covers the recommendations that the review note (docs/REVIEW.md) and the TRL 3 documents made after that: the two suggestions in the TRL 2 review note and the recommended next step in the TRL 3 review note, which proposed design responses to the at-risk requirements R9 (cell temperature) and R10 (installation force). Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so any recommendation that needs building, testing or measuring is recorded as decided but on hold.

■ Decision

Table 1. Newly decided items.

#	Item	Decision	What changed in the repo
A1	Slot tool for firm soil (TRL 2 review, suggestion; RMS-CAL-001 v0.1, section G)	Decided by Amish, 2026-09-25: go with recommendation. A tool pre-cuts the fin path. At TRL 3 it is a 32 x 8 mm mild steel flat bar, 650 mm, pointed at the EC tip depth, with a bolted T-handle and a printed depth-stop collar; steel rather than printed, because a printed blade would not survive firm soil.	cad/src/model.py adds build_slot_tool() and exports slot-tool.step and .stl ; BOM line 15 (\$10.00, one per set); RMS-CAL-001 v0.2 section G: push force 517 N to 376 N in moist loam and 1,778 N to 1,216 N in firm dry loam; R10 stays at risk; RMS-DWG-001 Rev P1 to P2 (slot tool note); RMS-PRC-001 v0.4 (component 15, design choice, safety).
A2	Deeper cell position for R9 (TRL 3 review, recommended next step)	Decided by Amish, 2026-09-25: go with recommendation. The deeper cell was checked by calculation.	RMS-CAL-001 v0.2 [F4]: 40 °C needs the cell center about 163 mm deep, where the upper probe sits; the deepest center that fits is about 65 mm (46.5 °C against 46.8 °C). The cell stays at 62 mm; no geometry change. R9 stays at risk and the remaining choice becomes P1 below.
A3	Pilot set now includes the slot tool (follows from A1)	Decided by Amish, 2026-09-25: go with recommendation. budget_usd stays \$300.	R12 restated so that the \$300 pilot set covers three stakes, one gateway and the slot tool; pilot set \$256.50 to \$266.50, headroom \$43.50 to \$33.50 (RMS-REQ-001 v0.4; bom/bom-notes.md ; concept blueprint key figures).
A4	Calibration note for the two-point field calibration (TRL 2 review, suggestion)	Decided by Amish, 2026-09-25: go with recommendation, but on hold. The EC cell constant is already calculated in RMS-CAL-001 section D; the field moisture calibration needs gravimetric samples and KCl standards, which is TRL 4 work.	None. On hold because TRL 4 is on hold by Amish's instruction.

Table 2. Items still open.

#	Item	Status
O1	First users and region for co-design (market garden, orchard, research farm or extension program), which also sets EU868 or US915. No recommendation was made.	Decided by Amish on 2026-10-02 as recommended in RMS-DEC-001: a US university extension program with a research farm, on US915, with Texas A&M AgriLife Extension as the first candidate to approach
P1	Response to the cell temperature in bare, hot soil (R9), raised by the check in A2. Option A: a printed shade skirt around the head to shade the soil over the cell (adds a part near machinery; its effect needs measuring). Option B: restate the R9 cell limit to the chosen cell's rated discharge and storage range, taken from its datasheet, while charging stays blocked above 45 °C. Recommendation: B, because the charger already blocks charging in the hot hours and B adds no part; confirm the cell's rated range before adopting it. This was a new proposal made after Amish's acceptance.	Decided by Amish on 2026-10-02 as recommended in RMS-DEC-001: option B, provided the cell's datasheet rates discharge to 55 °C or more; otherwise option A, the shade skirt

■ Consequences

- `project.yaml`: `budget_usd` stays \$300; pitch and problem unchanged; `trl: 3` and `trl_target: 3`.
- RMS-PRC-001 and RMS-REQ-001 move to v0.4, RMS-CAL-001 to v0.2, and drawing RMS-DWG-001 to Rev P2. RMS-PRB-001 is unchanged by these decisions.
- Requirement status is unchanged in count: 8 met, 5 at risk (R1, R5, R9, R10, R14), 1 not verifiable at TRL 3 (R8), none not met. R10 improves in moist soil but stays at risk in firm dry soil.
- No cross-repo action follows from these items.
- Nothing here moves the design toward TRL 4. A timed installation trial, the field calibration and any build of the slot tool are TRL 4 work and on hold.